Abishek Kamalarajan

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Abishek Kamalarajan Abishek-Kamalarajan Portfolio

PROFESSIONAL SUMMARY

Final-year Surveying Sciences undergraduate at SUSL, specializing in GIS, with a parallel Diploma in Information Technology at UCSC. Passionate about applying geospatial tools and Python-based workflows to real-world problems in urban planning and environmental analysis. Hands-on experience with ArcGIS, QGIS, and Google Earth Engine through academic projects and undergraduate research.

TECHNICAL SKILLS

GIS & Spatial:	QGIS, ArcGIS Pro, ArcMAP
Programming:	Python, SQL, JavaScript, HTML, CSS
Tools:	AutoCAD, Google Earth Engine, PyCharm, Excel

UNDERGRADUATE RESEARCH

Integrated Land-Use Transport Interaction (LUTI) Model for Sustainable Urban Growth in Kandy — *Python, AequilibraE, UrbanSim, PyCharm* Link

August 2025 – Present

- Designed and implemented a macroscopic LUTI model utilizing **Python**, **AequilibraE**, and **UrbanSim** to simulate and analyze multi-modal transportation scenarios across the Kandy district.
- Engineered robust geospatial workflows and spatial algorithms within **PyCharm** to evaluate urbanization-driven trends and sustainable development metrics.
- Directed comprehensive data acquisition, successfully sourcing and integrating 2024 transportation datasets from key national agencies (RDA, NTC, and Census Department).

FIELD EXPERIENCE

Survey Camp — HSL (High Flood Level) Demarcation Survey
Samanalawewa Reservoir *Academic Project - 2024*

- Part of a twelve-member survey team responsible for GNSS data collection, level transfer from benchmarks, and High Flood Level (HFL) peg placement along the Samanalawewa Reservoir reach for flood-risk zone management
- Operated GNSS receiver and Auto Level for benchmark transfer; recorded water-level data at cross-sections; processed observations in AutoCAD to map flood extent; co-authored the field camp technical report
- Successfully demarcated 464.700 m MSL of HFL boundary; field camp report graded A by supervising faculty

Engineering Survey
Department of Surveying & Geodesy, SUSL *Academic Project - 2024*

- Accountable for setting out centre-line pegs, calculating earthwork volumes, and producing longitudinal and cross-sectional profiles for a simulated road construction project
- Used Total Station for alignment setting out and Auto Level for levelling runs; computed cut-and-fill volumes and generated profile drawings in AutoCAD Civil 3D

Moderate Block Surveying
Department of Surveying & Geodesy, SUSL *Academic Project - 2024*

- Responsible for planning and executing a complete block survey using a Total Station instrument within a two-day field window
- Set up control network, conducted traverse measurements, recorded horizontal and vertical angles, processed raw data in AutoCAD Civil 3D, and produced a scaled plan

EDUCATION

B.Sc. Honours in Surveying Sciences — GIS Specialization October 2022 – Present
Sabaragamuwa University of Sri Lanka (SUSL) *Ratnapura, Sri Lanka*

Higher Diploma in Information Technology (HDIT) October 2025 – Present
University of Colombo School of Computing (UCSC) *Colombo, Sri Lanka*

Diploma in Information Technology (DIT) October 2024 – July 2025
University of Colombo School of Computing (UCSC) *Colombo, Sri Lanka*

G.C.E. Advanced Level — Physical Science 2020
Highlands Central College *Hatton, Sri Lanka*

EXTRA READINGS

- ArcGIS and ArcGIS Pro Fundamentals — Esri
- 100 Days of Code: The Complete Python Pro Bootcamp — Udemy
- Python Programming — Open Learning Platform, University of Moratuwa

MEMBERSHIPS

- International Society of Photogrammetry & Remote Sensing (ISPRS) Student Consortium
  Individual Member

LANGUAGES

English:	Professional Working Proficiency
Tamil:	Native Proficiency
Sinhala:	Limited Working Proficiency

REFERENCES

Professor RMKGSPB Koswatte
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Professor PGRNI Pussella
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